

LOVELY PROFESSIONAL UNIVERSITY

Course Code: CSE402 **Course Title:** Modern AI Systems & Agentic Orchestration

Credits: 4 (L:3, T:0, P:2) **Department:** Computer Science & Engineering

Course Outcomes (COs)

- **CO1:** Understand the foundations of Large Language Models and prompting paradigms.
- **CO2:** Apply single-agent design patterns (ReAct, Planning) to code execution tasks.
- **CO3:** Build multi-agent communication networks for collaborative software design.
- **CO4:** Set up CI/CD workflows and dockerized testing environments for agent systems.
- **CO5:** Audit applications using Bloom's Taxonomy cognitive standards and monitoring tools.

Syllabus Units Breakdown

Unit	Title & Description	Topics Covered
Unit 1	Basics of LLMs & API Drivers	Transformers, tokenizations, system instructions, temperature, Gemini/Groq APIs.
Unit 2	Prompt Engineering & RAG	Few-shot templates, Chain of Thought, vector databases, Semantic Search, chunking.
Unit 3	Autonomous Agent Loops	ReAct loop architectures, tool binding parameters, execution loops, state variables.
Unit 4	Multi-Agent Collaboration	GroupChat orchestrators, AutoGen communication models, reviewer-critic patterns.
Unit 5	CI/CD Deployment & Telemetry	GitHub Actions workflows, Docker containers, Prometheus dashboards, Sentry tracking.

Suggested Textbooks & Readings

1. *Introduction to Agentic Software Engineering*, LPU Academic Press, 2026.
2. *Large Language Models in Action*, Manning Publications, 2025.